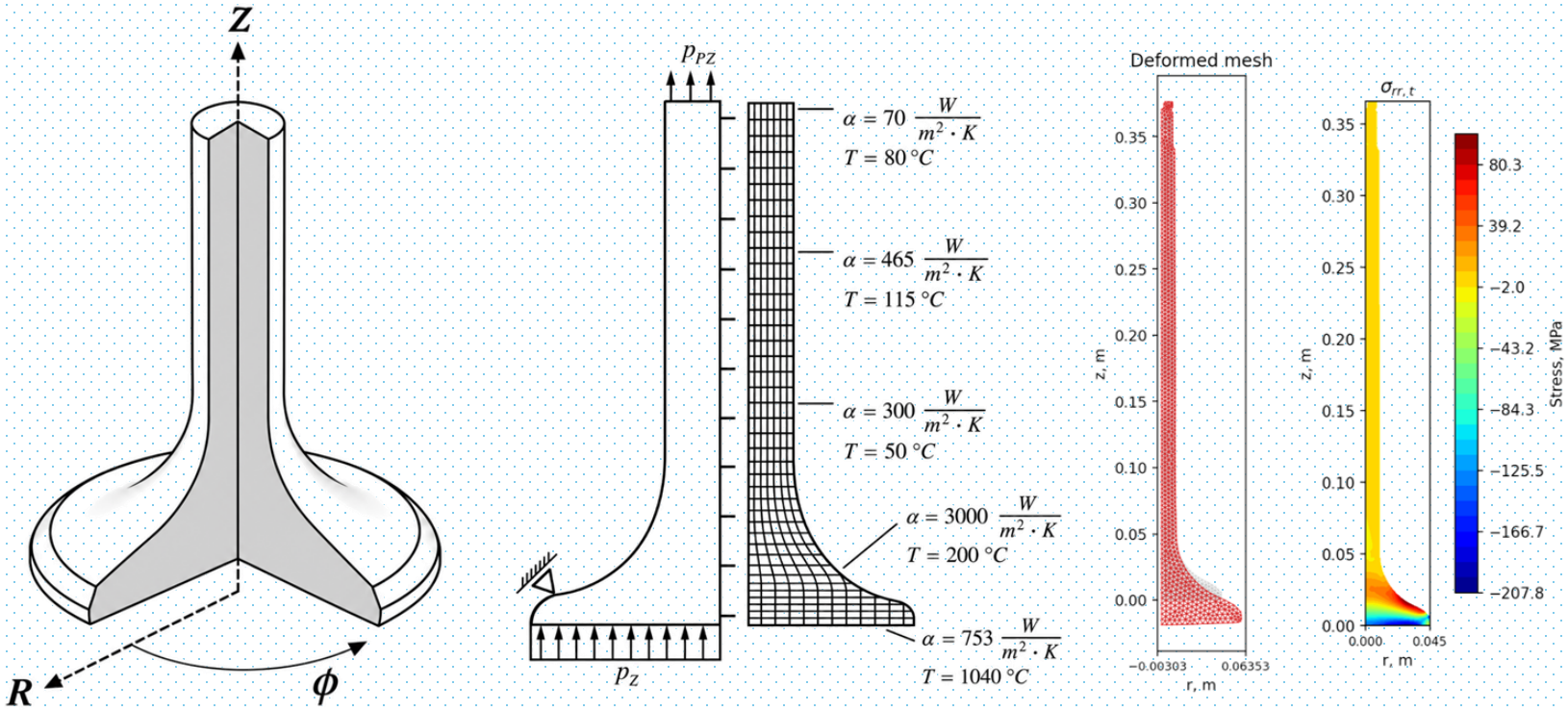
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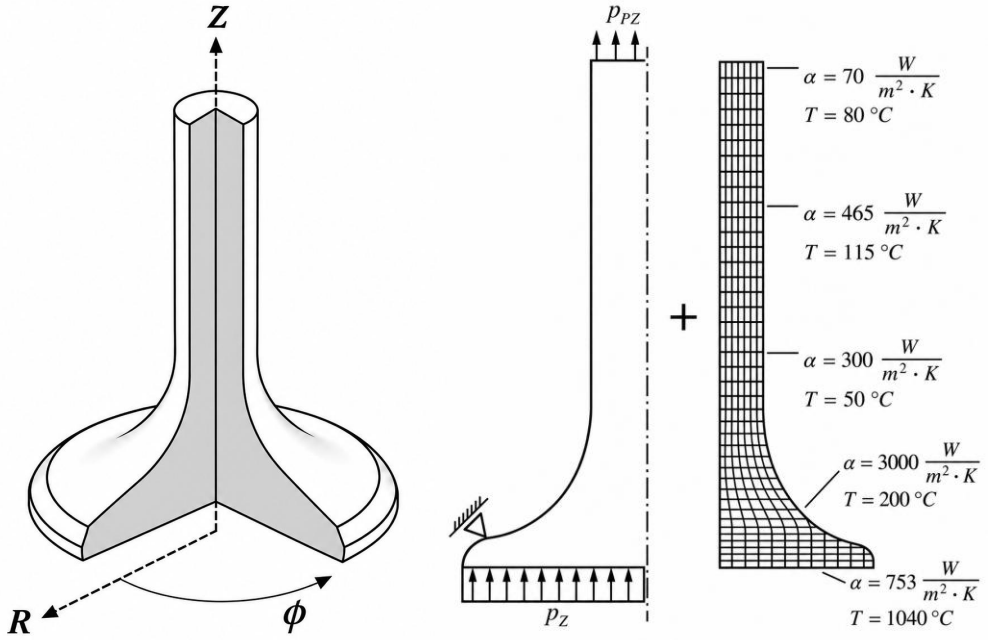
Axisymmetric Finite Element Analysis of the Thermoelastic Response of Engine Components

Special Topics in Finite Element Modeling for IC engines



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Problem statement: two sequential stages



1. Axisymmetric mechanical stress and strain analysis

Given: body-of-revolution geometry; elastic properties E and ν;
body forces and surface tractions; displacement constraints.

$$u \longrightarrow \varepsilon,\ \sigma,\ R$$

Find displacements, strains, stresses and support reactions.

2. Axisymmetric thermoelastic analysis (including thermal strains)

Additional inputs: temperature T(r, z);
stress-free reference temperature; thermal expansion coefficient.
The heat-conduction problem is not solved here.

$$T \longrightarrow \varepsilon_{\text{th}} \longrightarrow u,\ \varepsilon_{\text{el}},\ \sigma,\ R$$

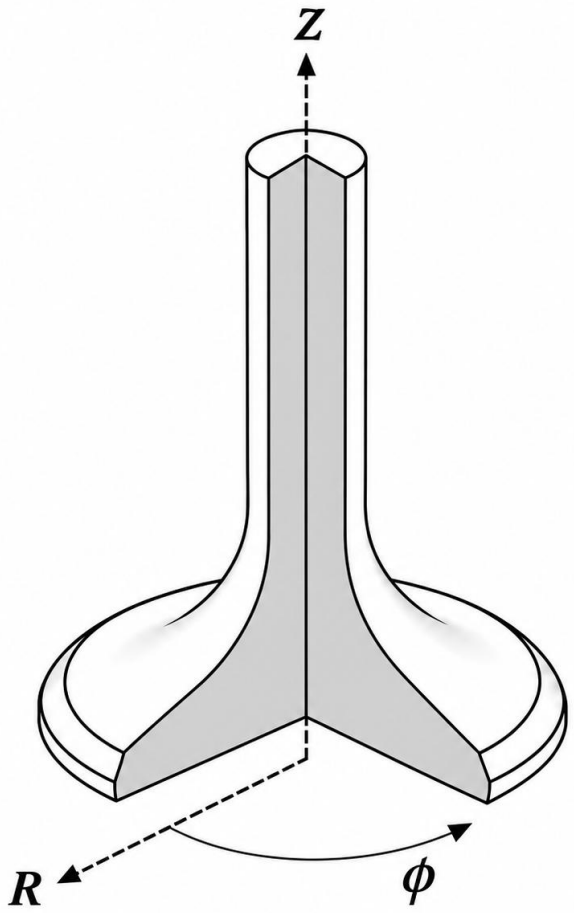
Obtain the updated mechanical response:
stress depends on the elastic strain.

Model assumptions

Static equilibrium. Small strains. Isotropic linear elasticity.
Mechanical loads do not change as the body deforms.

First determine the mechanical response, then account for the change caused by thermal strains.

Force balance in cylindrical coordinates



r: radial coordinate; φ: angle;
z: axial coordinate.

1. Force balance for an arbitrary volume

$$\int_V \rho\,\ddot{\boldsymbol{u}}\,dV = \int_{\partial V} \boldsymbol{\sigma}\boldsymbol{n}\,dS + \int_V \boldsymbol{b}\,dV$$

$$\boldsymbol{u} = (u_r, u_\varphi, u_z)^{\mathsf{T}}, \qquad dV = r\,dr\,d\varphi\,dz$$

u: displacement; ρ: density, kg/m³; b: body force per unit volume, N/m³.
V: volume; ∂V: its boundary; two dots above u denote acceleration.

2. Local form

$$\rho\,\ddot{\boldsymbol{u}} = \nabla\cdot\boldsymbol{\sigma} + \boldsymbol{b}$$

3. Surface traction

$$\boldsymbol{p} = \boldsymbol{\sigma}\boldsymbol{n}$$

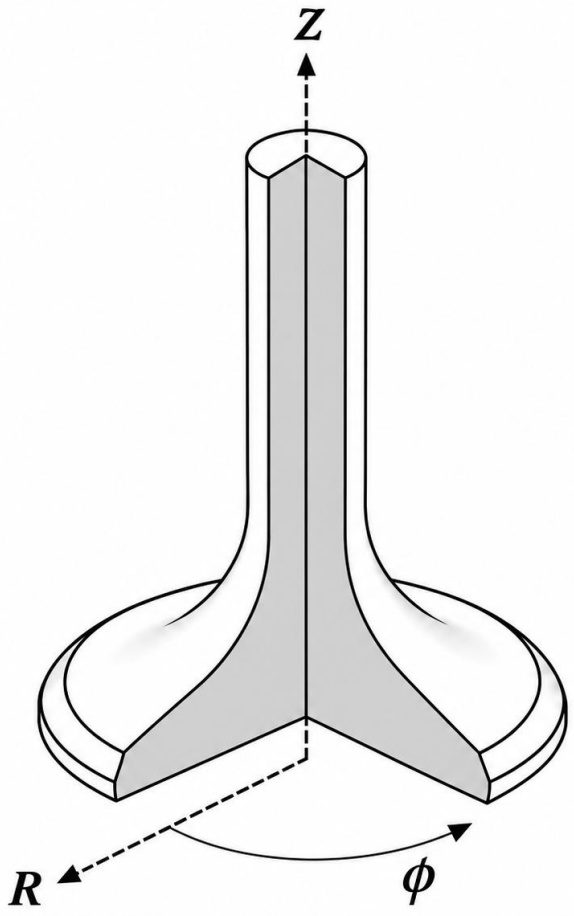
n: outward unit normal.
p: surface traction vector, N/m².

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{rr} & \tau_{r\varphi} & \tau_{rz} \\ \tau_{r\varphi} & \sigma_{\varphi\varphi} & \tau_{\varphi z} \\ \tau_{rz} & \tau_{\varphi z} & \sigma_{zz} \end{pmatrix}_{(r,\varphi,z)}$$

Component order: (r, φ, z).
σ: stress, Pa.

Force balance relates inertia, surface tractions and body forces. We next consider the static case.

Static equilibrium in cylindrical coordinates



r: radial coordinate; φ: angle;
z: axial coordinate.

1. Acceleration and the inertia term vanish

$$\ddot{\boldsymbol{u}} = 0 \quad \Longrightarrow \quad \rho \underbrace{\ddot{\boldsymbol{u}}}_{0} = \nabla \cdot \boldsymbol{\sigma} + \boldsymbol{b} = 0$$

In a static analysis, inertia does not enter the force balance.

2. Equilibrium in three directions

Radial r

$$\frac{\partial \sigma_{rr}}{\partial r} + \frac{1}{r}\frac{\partial \tau_{r\varphi}}{\partial \varphi} + \frac{\partial \tau_{rz}}{\partial z} + \frac{\sigma_{rr} - \sigma_{\varphi\varphi}}{r} + b_r = 0$$

Hoop φ

$$\frac{\partial \tau_{r\varphi}}{\partial r} + \frac{1}{r}\frac{\partial \sigma_{\varphi\varphi}}{\partial \varphi} + \frac{\partial \tau_{\varphi z}}{\partial z} + \frac{2\tau_{r\varphi}}{r} + b_\varphi = 0$$

Axial z

$$\frac{\partial \tau_{rz}}{\partial r} + \frac{1}{r}\frac{\partial \tau_{\varphi z}}{\partial \varphi} + \frac{\partial \sigma_{zz}}{\partial z} + \frac{\tau_{rz}}{r} + b_z = 0$$

Retain all stress components and dependence on φ.

Axisymmetry is not yet assumed: all three directions and dependence on φ are retained.

Kinematics in cylindrical coordinates

1. Three displacement components and small strains

$$\boldsymbol{u} = u_r(r,\varphi,z)\boldsymbol{e}_r + u_\varphi(r,\varphi,z)\boldsymbol{e}_\varphi + u_z(r,\varphi,z)\boldsymbol{e}_z$$

$$\gamma_{ij} = 2\varepsilon_{ij} \quad (i \neq j), \qquad i,j \in \{r,\varphi,z\}$$

2. Normal strains

$$\varepsilon_{rr} = \frac{\partial u_r}{\partial r}$$

$$\varepsilon_{\varphi\varphi} = \frac{u_r}{r} + \frac{1}{r}\frac{\partial u_\varphi}{\partial \varphi}$$

$$\varepsilon_{zz} = \frac{\partial u_z}{\partial z}$$

3. Engineering shear strains

$$\gamma_{r\varphi} = \frac{\partial u_\varphi}{\partial r} - \frac{u_\varphi}{r} + \frac{1}{r}\frac{\partial u_r}{\partial \varphi}$$

$$\gamma_{rz} = \frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r}$$

$$\gamma_{\varphi z} = \frac{\partial u_\varphi}{\partial z} + \frac{1}{r}\frac{\partial u_z}{\partial \varphi}$$

Components may depend on r, φ and z. The 1/r terms account for the rotation of the radial and circumferential basis vectors as the angle φ changes.

Kinematics relates strains to displacements, independently of the loads and the constitutive law.

Three-dimensional isotropic Hooke's law

1. Linear relation between stress and small strain

$$\sigma_{ij} = 2G\varepsilon_{ij} + \lambda\left(\varepsilon_{rr} + \varepsilon_{\varphi\varphi} + \varepsilon_{zz}\right)\delta_{ij}$$

$$i, j \in \{r, \varphi, z\}, \qquad \delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

2. Elastic properties and Lamé parameters

$$G = \frac{E}{2(1+\nu)}, \qquad \lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}$$

E: Young's modulus; ν: Poisson's ratio.
λ: first Lamé parameter; G: shear modulus.

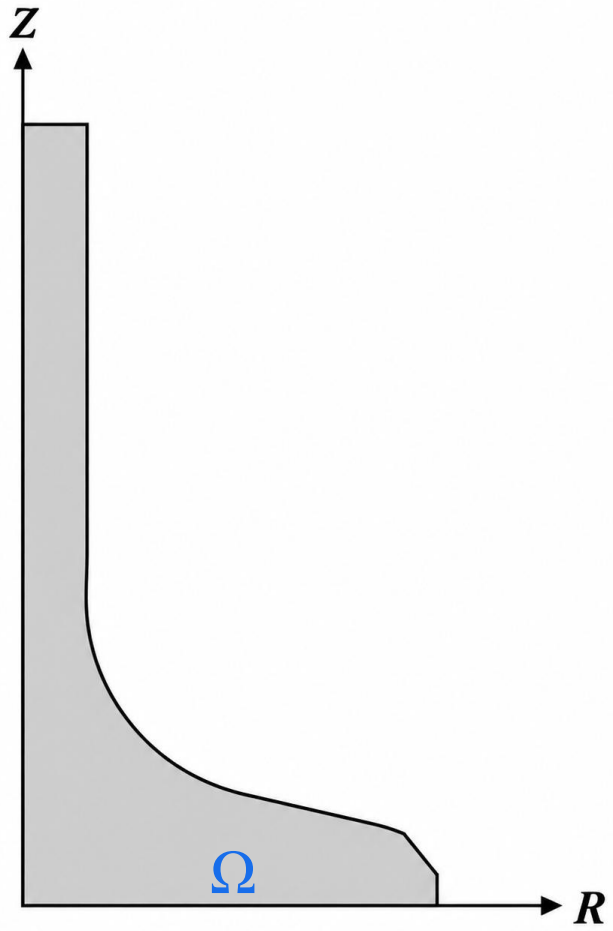
3. Shear stresses and engineering shear strains

$$\tau_{r\varphi} = 2G\varepsilon_{r\varphi} = G\gamma_{r\varphi} \qquad \tau_{rz} = 2G\varepsilon_{rz} = G\gamma_{rz} \qquad \tau_{\varphi z} = 2G\varepsilon_{\varphi z} = G\gamma_{\varphi z}$$

This mechanical problem has no thermal strains or other eigenstrains.

The constitutive law relates strain to stress. Axisymmetry has not yet been imposed.

Axisymmetry without torsion: the meridional domain



Ω: meridional domain,
$r \geq 0$.

1. Axisymmetry conditions

Geometry, material, loads and constraints are invariant under rotation about the z-axis.

$$\frac{\partial f}{\partial \varphi} = 0$$

f: a component of the displacement, strain or stress field expressed in cylindrical coordinates.

2. Additional assumption: no torsion

There are no circumferential loads.

$$u_\varphi = 0 \quad \Longrightarrow \quad \gamma_{r\varphi} = \gamma_{\varphi z} = 0, \quad \tau_{r\varphi} = \tau_{\varphi z} = 0$$

$$\boldsymbol{u}(r,z) = (u_r(r,z), u_z(r,z))^\mathsf{T}$$

3. Integrate over the angle from 0 to 2π

$$\int_0^{2\pi} r \, d\varphi = 2\pi r$$

$$\underbrace{2\pi r \, dA}_{\text{volume}}, \qquad \underbrace{2\pi r \, ds}_{\text{surface area}}$$

dA = dr dz; ds: arc-length element on the boundary of Ω.

Dependence on φ disappears, but circumferential strain and stress remain.

Axisymmetric strains and Hooke's law

1. Two displacements give four strain components

$$\varepsilon_{rr} = \frac{\partial u_r}{\partial r} \qquad \varepsilon_{zz} = \frac{\partial u_z}{\partial z} \qquad \varepsilon_{\varphi\varphi} = \frac{u_r}{r}$$

$$\gamma_{rz} = \frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} \qquad \varepsilon_{\varphi\varphi} = \frac{2\pi(r + u_r) - 2\pi r}{2\pi r} = \frac{u_r}{r}$$

2. Component order and reduced constitutive matrix

$$\boldsymbol{\varepsilon} = (\varepsilon_{rr}, \varepsilon_{zz}, \varepsilon_{\varphi\varphi}, \gamma_{rz})^{\mathsf{T}}, \quad \boldsymbol{\sigma} = (\sigma_{rr}, \sigma_{zz}, \sigma_{\varphi\varphi}, \tau_{rz})^{\mathsf{T}}$$

Component order: (r, φ, z)

$$\boldsymbol{\sigma} = \boldsymbol{D}\boldsymbol{\varepsilon}, \qquad \boldsymbol{D} = \begin{pmatrix} \lambda + 2G & \lambda & \lambda & 0 \\ \lambda & \lambda + 2G & \lambda & 0 \\ \lambda & \lambda & \lambda + 2G & 0 \\ 0 & 0 & 0 & G \end{pmatrix} \qquad \boldsymbol{\sigma}_{\text{tensor}} = \begin{pmatrix} \sigma_{rr} & 0 & \tau_{rz} \\ 0 & \sigma_{\varphi\varphi} & 0 \\ \tau_{rz} & 0 & \sigma_{zz} \end{pmatrix}$$

λ: first Lamé parameter; G: shear modulus. With engineering shear strain, $D_{44} = G$.

The model uses two spatial coordinates but retains circumferential strain and stress.

Equilibrium and traction boundary conditions

1. Two equilibrium equations in Ω

$$\frac{\partial \sigma_{rr}}{\partial r} + \frac{\partial \tau_{rz}}{\partial z} + \frac{\sigma_{rr} - \sigma_{\varphi\varphi}}{r} + b_r = 0$$

$$\frac{\partial \tau_{rz}}{\partial r} + \frac{\partial \sigma_{zz}}{\partial z} + \frac{\tau_{rz}}{r} + b_z = 0$$

Circumferential equilibrium is satisfied: circumferential loads and shear components involving φ vanish.

2. Prescribed surface tractions

$$\begin{pmatrix} \sigma_{rr} & \tau_{rz} \\ \tau_{rz} & \sigma_{zz} \end{pmatrix} \begin{pmatrix} n_r \\ n_z \end{pmatrix} = \begin{pmatrix} \bar{p}_r \\ \bar{p}_z \end{pmatrix} \quad \text{on } \Gamma_t$$

Γ_t: traction boundary of the meridional domain.
The force per unit area is prescribed.
n: outward unit normal.

3. Pressure

$$\bar{\boldsymbol{p}} = -p_n \boldsymbol{n}, \qquad p_n \geq 0$$

p_n: compressive pressure, Pa.
The minus sign gives the inward direction.
A general surface traction need not
be normal to the boundary.

Equilibrium holds within the meridional domain; prescribed tractions act on its boundary.

Displacement constraints and regularity at the axis

1. Prescribed displacement components

$$u_r = \bar{u}_r \quad \text{or} \quad u_z = \bar{u}_z \quad \text{on the corresponding part of } \Gamma_u$$

Γ_u: displacement boundary. Prescribed values may be zero or nonzero.

2. Ideal inclined frictionless support

$$u_n = n_r u_r + n_z u_z = 0, \qquad u_s = -n_z u_r + n_r u_z, \qquad p_s = 0$$

u_n and u_s: normal and tangential displacements; p_s: tangential traction.
This is a bilateral restraint. Unilateral contact requires a separate model.

3. Regularity conditions at r = 0

$$u_r(0, z) = 0, \qquad \left.\frac{\partial u_z}{\partial r}\right|_{r=0} = 0$$

$$\lim_{r \to 0} \frac{u_r}{r} = \left.\frac{\partial u_r}{\partial r}\right|_{r=0}, \qquad \varepsilon_{\varphi\varphi}(0, z) = \varepsilon_{rr}(0, z)$$

Constraints restrict motion. Axis regularity ensures a smooth field without imposing an axial restraint.

From equilibrium to minimum potential energy

1. Total potential energy of the mechanical system

In linear elastic statics with deformation-independent loads,
equilibrium minimizes the total potential energy.

$$\Pi[\boldsymbol{u}] = U - W_b - W_t$$

Elastic strain energy

$$U = \frac{1}{2}\int_\Omega \boldsymbol{\varepsilon}^\top \boldsymbol{D}\boldsymbol{\varepsilon}\, 2\pi r\, dA$$

Work of body forces

$$W_b = \int_\Omega \boldsymbol{u}^\top \boldsymbol{b}\, 2\pi r\, dA$$

Work of surface tractions

$$W_t = \int_{\Gamma_t} \boldsymbol{u}^\top \bar{\boldsymbol{p}}\, 2\pi r\, ds$$

2. Minimum over admissible displacements

$$\Pi[\boldsymbol{u}] \longrightarrow \min$$

Variations of prescribed displacement components vanish.
On a frictionless support, $\delta u_n = 0$.
The minimum is unique once rigid-body modes are removed.

3. Nodal approximation and algebraic equations

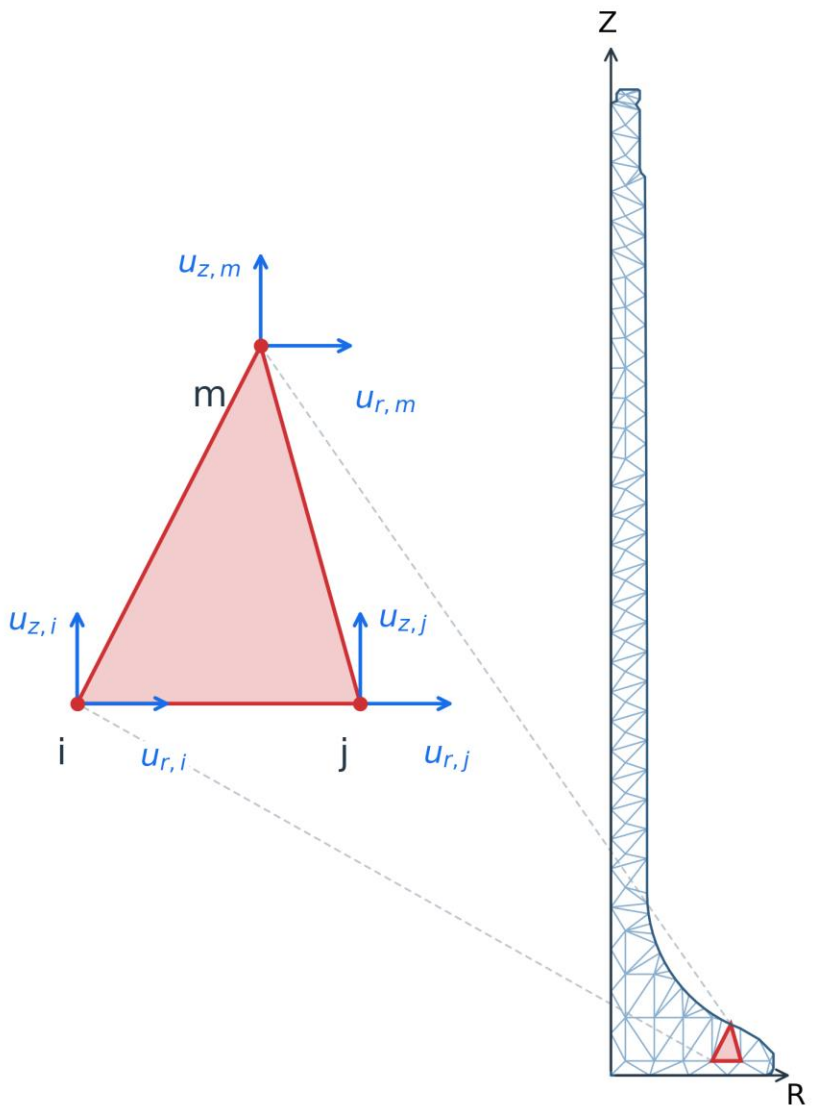
For independent unknown nodal displacement components after enforcing the constraints

$$\frac{\partial \Pi_h}{\partial u_{r,i}} = 0, \qquad \frac{\partial \Pi_h}{\partial u_{z,i}} = 0$$

Π_h: functional of the nodal displacements.
i: node number; nodal displacements
are radial (r) and axial (z).

We derive each contribution to the functional before obtaining the coefficients of the algebraic system.

Finite element model of the meridional domain



1. Triangulate the meridional domain

Rotating each triangle about the z-axis
generates the volume of an axisymmetric element.

$$V_e = \int_{\Omega_e} 2\pi r \, dA$$

Ω_e: triangle in the meridional plane.
V_e: corresponding volume of revolution.

2. Two unknowns at each node

$$\boldsymbol{u}_e = \left(u_{r,i}, u_{z,i}, u_{r,j}, u_{z,j}, u_{r,m}, u_{z,m}\right)^{\mathsf{T}}$$

u_e: vector of six element nodal displacements.
Each node i, j, m has a radial and an axial displacement.

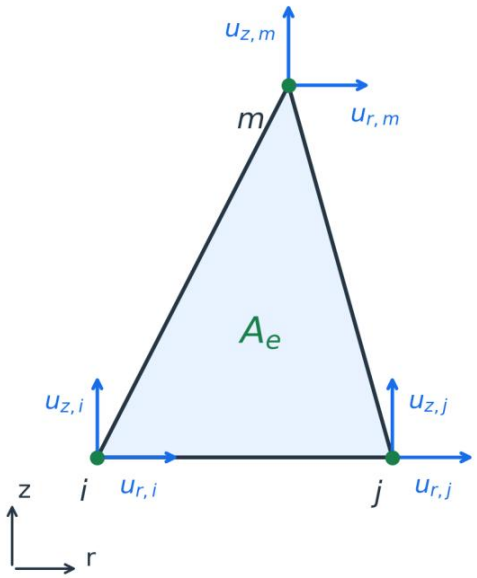
3. Assemble the global functional

$$\Pi_h = \sum_{e=1}^{n_e} \Pi_e$$

n_e: number of elements. Adjacent elements
share the same displacements at common nodes.

The unknowns are nodal displacements; the field inside each element is expressed in terms of them.

Displacements within a triangular element



A_e: element area.
Nodal coordinates: r and z with indices i, j, m.
The order i → j → m is counterclockwise.

1. Linear displacement approximation

$$u_r = \alpha_0 + \alpha_r r + \alpha_z z, \qquad u_z = \beta_0 + \beta_r r + \beta_z z.$$

2. Apply nodal values: three equations each for α and β

$$\begin{cases} u_{r,i} = \alpha_0 + \alpha_r r_i + \alpha_z z_i, \\ u_{r,j} = \alpha_0 + \alpha_r r_j + \alpha_z z_j, \\ u_{r,m} = \alpha_0 + \alpha_r r_m + \alpha_z z_m. \end{cases} \qquad \begin{cases} u_{z,i} = \beta_0 + \beta_r r_i + \beta_z z_i, \\ u_{z,j} = \beta_0 + \beta_r r_j + \beta_z z_j, \\ u_{z,m} = \beta_0 + \beta_r r_m + \beta_z z_m. \end{cases}$$

3. Shape functions and their coefficients

$$N_\ell(r,z) = \frac{a_\ell + b_\ell r + c_\ell z}{2A_e}, \qquad \ell = i, j, m$$

$$a_\ell = r_p z_q - r_q z_p, \qquad b_\ell = z_p - z_q, \qquad c_\ell = r_q - r_p$$

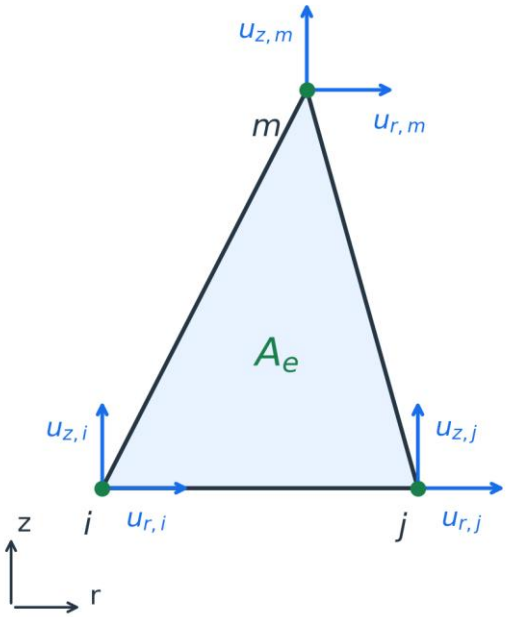
$$(\ell, p, q) = (i, j, m),\ (j, m, i),\ (m, i, j)$$

p and q are the other nodes in the stated cyclic order.

$$u_k = N_i u_{k,i} + N_j u_{k,j} + N_m u_{k,m}, \qquad k = r \text{ or } z$$

Shape functions N_i, N_j, N_m express the linear displacement field in terms of nodal values.

Displacement interpolation matrix



i, j, m: element nodes.
A_e: triangle area.
r and z: point coordinates.

1. Order of element nodal displacements

$$\boldsymbol{u}_e = \left(u_{r,i}, u_{z,i}, u_{r,j}, u_{z,j}, u_{r,m}, u_{z,m}\right)^{\mathsf{T}}$$

2. Write interpolation in matrix form

$$\boldsymbol{u}(r,z) = \begin{pmatrix} u_r(r,z) \\ u_z(r,z) \end{pmatrix} = \boldsymbol{N}(r,z)\boldsymbol{u}_e$$

$$\boldsymbol{N}(r,z) = \begin{pmatrix} N_i & 0 & N_j & 0 & N_m & 0 \\ 0 & N_i & 0 & N_j & 0 & N_m \end{pmatrix}$$

N maps nodal values to the displacement field and is used to evaluate the work of the loads.

3. Shape-function derivatives

$$\frac{\partial N_\ell}{\partial r} = \frac{b_\ell}{2A_e}, \qquad \frac{\partial N_\ell}{\partial z} = \frac{c_\ell}{2A_e}, \qquad \ell = i, j, m$$

Matrix N interpolates the displacement field, from which the strains are then obtained.

Strains within a triangular element

The derivatives of linear shape functions are constant

$$\frac{\partial N_\ell}{\partial r} = \frac{b_\ell}{2A_e}, \qquad \frac{\partial N_\ell}{\partial z} = \frac{c_\ell}{2A_e}, \qquad \ell = i, j, m$$

1. Radial

$$\varepsilon_{rr} = \frac{\partial u_r}{\partial r} = \frac{b_i u_{r,i} + b_j u_{r,j} + b_m u_{r,m}}{2A_e}$$

2. Axial

$$\varepsilon_{zz} = \frac{\partial u_z}{\partial z} = \frac{c_i u_{z,i} + c_j u_{z,j} + c_m u_{z,m}}{2A_e}$$

3. Hoop

$$\varepsilon_{\varphi\varphi} = \frac{u_r}{r} = \frac{N_i u_{r,i} + N_j u_{r,j} + N_m u_{r,m}}{r}$$

4. Shear

$$\gamma_{rz} = \frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} = \frac{c_i u_{r,i} + c_j u_{r,j} + c_m u_{r,m} + b_i u_{z,i} + b_j u_{z,j} + b_m u_{z,m}}{2A_e}$$

Radial, axial and shear strains are constant; circumferential strain generally varies.

Strain–displacement matrix

1. Component order: displacements and strains

$$\boldsymbol{u}_e = (u_{r,i}, u_{z,i}, u_{r,j}, u_{z,j}, u_{r,m}, u_{z,m})^{\mathsf{T}} \qquad \boldsymbol{\varepsilon} = (\varepsilon_{rr}, \varepsilon_{zz}, \varepsilon_{\varphi\varphi}, \gamma_{rz})^{\mathsf{T}}$$

2. Collect coefficients of the four kinematic relations

$$\boldsymbol{B}(r,z) = \begin{pmatrix} \frac{b_i}{2A_e} & 0 & \frac{b_j}{2A_e} & 0 & \frac{b_m}{2A_e} & 0 \\ 0 & \frac{c_i}{2A_e} & 0 & \frac{c_j}{2A_e} & 0 & \frac{c_m}{2A_e} \\ \frac{N_i}{r} & 0 & \frac{N_j}{r} & 0 & \frac{N_m}{r} & 0 \\ \frac{c_i}{2A_e} & \frac{b_i}{2A_e} & \frac{c_j}{2A_e} & \frac{b_j}{2A_e} & \frac{c_m}{2A_e} & \frac{b_m}{2A_e} \end{pmatrix}$$

3. Obtain the strain and stress fields

$$\boldsymbol{\varepsilon} = \boldsymbol{B}\boldsymbol{u}_e \qquad \boldsymbol{\sigma} = \boldsymbol{D}\boldsymbol{\varepsilon} = \boldsymbol{D}\boldsymbol{B}\boldsymbol{u}_e$$

B: strain–displacement matrix; D: the previously defined constitutive matrix.

The circumferential row makes B dependent on position within the element.

Elastic strain energy and element stiffness matrix

1. Initial contribution: elastic strain energy

$$U_e = \frac{1}{2}\int_{\Omega_e} \boldsymbol{\varepsilon}^{\mathsf{T}} \boldsymbol{D}\boldsymbol{\varepsilon}\, 2\pi r\, dA$$

2. Substitute strains in terms of nodal displacements

$$U_e = \frac{1}{2}\int_{\Omega_e} (\boldsymbol{B}\boldsymbol{u}_e)^{\mathsf{T}} \boldsymbol{D}(\boldsymbol{B}\boldsymbol{u}_e)\, 2\pi r\, dA = \frac{1}{2}\boldsymbol{u}_e^{\mathsf{T}} \left[\int_{\Omega_e} \boldsymbol{B}^{\mathsf{T}}\boldsymbol{D}\boldsymbol{B}\, 2\pi r\, dA\right] \boldsymbol{u}_e = \frac{1}{2}\boldsymbol{u}_e^{\mathsf{T}}\boldsymbol{K}_e\boldsymbol{u}_e$$

The nodal vector u_e is independent of r and z. Matrix B varies spatially and remains inside the integral.

3. Element stiffness matrix and its approximate evaluation

$$\boldsymbol{K}_e = \int_{\Omega_e} \boldsymbol{B}^{\mathsf{T}}\boldsymbol{D}\boldsymbol{B}\, 2\pi r\, dA \;\approx\; 2\pi r_c A_e \boldsymbol{B}_c^{\mathsf{T}}\boldsymbol{D}\boldsymbol{B}_c$$

$$\boldsymbol{K}_e = \begin{bmatrix} k_{11} & k_{12} & k_{13} & k_{14} & k_{15} & k_{16} \\ k_{21} & k_{22} & k_{23} & k_{24} & k_{25} & k_{26} \\ k_{31} & k_{32} & k_{33} & k_{34} & k_{35} & k_{36} \\ k_{41} & k_{42} & k_{43} & k_{44} & k_{45} & k_{46} \\ k_{51} & k_{52} & k_{53} & k_{54} & k_{55} & k_{56} \\ k_{61} & k_{62} & k_{63} & k_{64} & k_{65} & k_{66} \end{bmatrix}$$

$$r_c = \frac{r_i + r_j + r_m}{3}, \qquad z_c = \frac{z_i + z_j + z_m}{3};$$

$\boldsymbol{B}_c = \boldsymbol{B}(r_c, z_c)$, A_e: meridional triangle area.

The elastic contribution is now derived. The work of body forces and surface tractions is added separately.

Body-force work and the nodal load vector

1. Initial contribution: work of body forces

$$\boldsymbol{b} = (b_r, b_z)^{\mathsf{T}}$$

Body force per unit volume, N/m³.

$$W_{b,e} = \int_{\Omega_e} \boldsymbol{u}^{\mathsf{T}} \boldsymbol{b}\, 2\pi r\, dA = \int_{\Omega_e} (u_r b_r + u_z b_z)\, 2\pi r\, dA$$

2. Substitute the displacement field

$$W_{b,e} = \int_{\Omega_e} (\boldsymbol{N}\boldsymbol{u}_e)^{\mathsf{T}} \boldsymbol{b}\, 2\pi r\, dA$$

$$W_{b,e} = \boldsymbol{u}_e^{\mathsf{T}} \left[\int_{\Omega_e} \boldsymbol{N}^{\mathsf{T}} \boldsymbol{b}\, 2\pi r\, dA \right]$$

$$W_{b,e} = \boldsymbol{u}_e^{\mathsf{T}} \boldsymbol{f}_e^{b}$$

3. Nodal forces, N

$$\boldsymbol{f}_e^{b} = \int_{\Omega_e} \begin{pmatrix} N_i b_r \\ N_i b_z \\ N_j b_r \\ N_j b_z \\ N_m b_r \\ N_m b_z \end{pmatrix} 2\pi r\, dA$$

For constant b: k = r or z; use cyclic permutations for nodes j and m.

$$f_{k,i}^{b} = 2\pi b_k \int_{\Omega_e} r N_i\, dA = \frac{\pi A_e}{6} (2r_i + r_j + r_m) b_k$$

A distributed body force produces a consistent nodal load vector.

Work of surface tractions on an element edge

1. Work and nodal load vector

$\bar{p}$: surface traction vector, N/m².
$\Gamma_{t,e}$: loaded element boundary.
Traction need not be normal to the boundary.

$$W_{t,e} = \int_{\Gamma_{t,e}} \boldsymbol{u}^{\mathsf{T}} \bar{\boldsymbol{p}}\, 2\pi r\, ds$$

$$W_{t,e} = \int_{\Gamma_{t,e}} (\boldsymbol{N}\boldsymbol{u}_e)^{\mathsf{T}} \bar{\boldsymbol{p}}\, 2\pi r\, ds$$

$$W_{t,e} = \boldsymbol{u}_e^{\mathsf{T}} \underbrace{\int_{\Gamma_{t,e}} \boldsymbol{N}^{\mathsf{T}} \bar{\boldsymbol{p}}\, 2\pi r\, ds}_{\boldsymbol{f}_e^t}$$

$$W_{t,e} = \boldsymbol{u}_e^{\mathsf{T}} \boldsymbol{f}_e^t$$

The vector u_e is independent of r and z and can be taken outside the integral.

2. Constant traction on edge ij

L: edge length; s: distance from node i.
$0 \le s \le L$; k = r or z.

$$r(s) = r_i\left(1 - \frac{s}{L}\right) + r_j \frac{s}{L}$$

$$N_i = 1 - \frac{s}{L}, \qquad N_j = \frac{s}{L}, \qquad N_m = 0$$

$$f_{k,i}^t = 2\pi \bar{p}_k \int_0^L r(s)\left(1 - \frac{s}{L}\right) ds = \frac{\pi L}{3}(2r_i + r_j)\bar{p}_k,$$

$$f_{k,j}^t = 2\pi \bar{p}_k \int_0^L r(s)\frac{s}{L}\, ds = \frac{\pi L}{3}(r_i + 2r_j)\bar{p}_k.$$

$$f_{r,m}^t = f_{z,m}^t = 0$$

These formulas are exact for constant $\bar{p}$ on a straight element edge.

The factor 2πr accounts for the surface of revolution. An edge on the axis r = 0 has zero contribution.

Global functional and algebraic system

1. Combine strain energy and both load-work contributions

$$\Pi_h = \sum\nolimits_e \Pi_e, \qquad \Pi_e = \frac{1}{2}\boldsymbol{u}_e^{\mathsf{T}}\boldsymbol{K}_e\boldsymbol{u}_e - \boldsymbol{u}_e^{\mathsf{T}}(\boldsymbol{f}_e^b + \boldsymbol{f}_e^t)$$

2. Assemble by global indices; initialize K = 0 and F = 0

$$(I_1, I_2, I_3, I_4, I_5, I_6) = (2i-1, 2i, 2j-1, 2j, 2m-1, 2m)$$

$$K_{I_a I_b} \leftarrow K_{I_a I_b} + (K_e)_{ab}, \qquad F_{I_a} \leftarrow F_{I_a} + (f_e^b + f_e^t)_a$$

I_a: global index of local component a; a, b = 1,…,6.
Nodes i, j, m are globally numbered; radial and axial displacements alternate in u.

3. Minimize the assembled functional

Apply these conditions only to independent unknowns after enforcing the constraints; s is the node number.

$$\frac{\partial \Pi_h}{\partial u_{r,s}} = (\boldsymbol{K}\boldsymbol{u} - \boldsymbol{F})_{2s-1} = 0 \qquad \frac{\partial \Pi_h}{\partial u_{z,s}} = (\boldsymbol{K}\boldsymbol{u} - \boldsymbol{F})_{2s} = 0$$

$$\boldsymbol{K}\boldsymbol{u} = \boldsymbol{F} + \boldsymbol{R}$$

F: mechanical nodal load vector; R: support reactions. In independent unconstrained directions, R = 0.

Minimize the functional of the entire model, not that of each element separately.

Imposing prescribed nodal displacements

1. Zero prescribed displacement components: $u_1 = u_6 = 0$

$$u_1 = u_{r,1} = 0, \qquad u_6 = u_{z,3} = 0$$

K_{11}	K_{12}	K_{13}	K_{14}	K_{15}	K_{16}	0	$F_1 + R_1$
K_{21}	K_{22}	K_{23}	K_{24}	K_{25}	K_{26}	u_2	F_2
K_{31}	K_{32}	K_{33}	K_{34}	K_{35}	K_{36}	u_3	F_3
K_{41}	K_{42}	K_{43}	K_{44}	K_{45}	K_{46}	u_4	F_4
K_{51}	K_{52}	K_{53}	K_{54}	K_{55}	K_{56}	u_5	F_5
K_{61}	K_{62}	K_{63}	K_{64}	K_{65}	K_{66}	0	$F_6 + R_6$

$=$

J = {2, 3, 4, 5}

Retain the green rows and columns

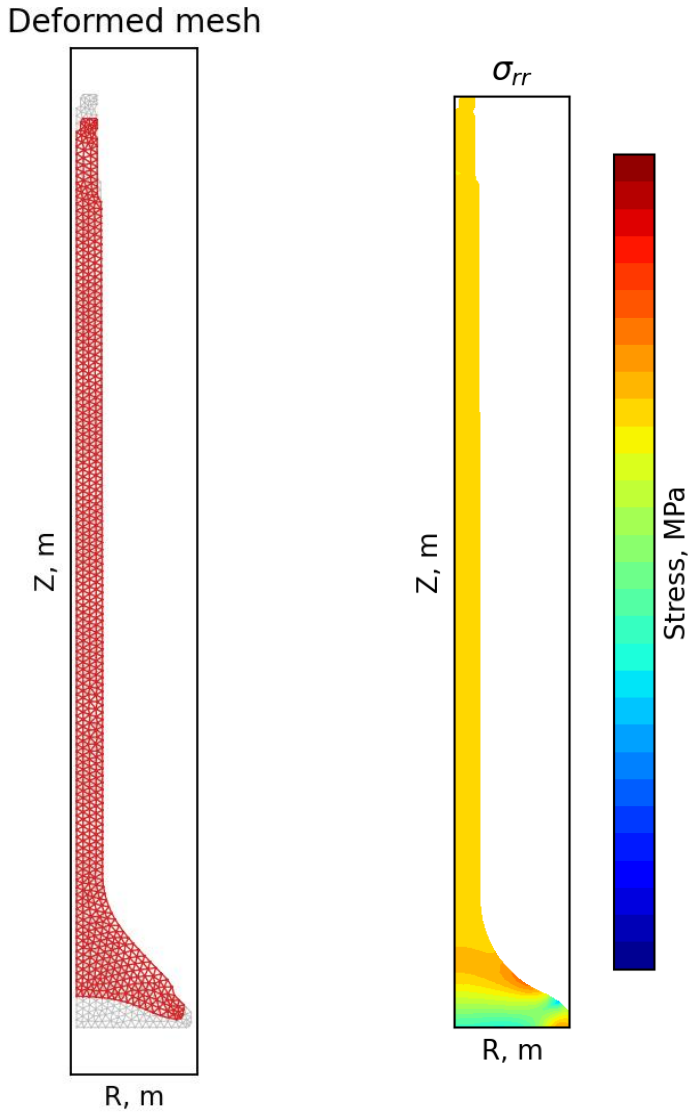
2. Arbitrary prescribed values

$$\sum_{b \in J} K_{ab} u_b = F_a - \sum_{c \in C} K_{ac} \bar{u}_c, \qquad a \in J$$

C indexes prescribed components; J indexes unknown components. An overbar denotes a prescribed value.
For an inclined support, first resolve the displacement into normal and tangential components.

For nonzero prescribed displacements, move the known contribution to the right-hand side before elimination.

Solving the system and recovering displacement, strain and stress fields



1. Solution and full displacement vector

Use a linear solver; do not explicitly invert the matrix.

$$u_a = \begin{cases} \text{computed value}, & a \in J, \\ \bar{u}_a, & a \in C, \end{cases} \qquad \boldsymbol{R} = \boldsymbol{K}\boldsymbol{u} - \boldsymbol{F}$$

Insert prescribed values at constrained components.
Use the original K and F, before elimination.

2. Inclined frictionless support

$$u_{r,i} = -n_z u_{s,i}, \qquad u_{z,i} = n_r u_{s,i}$$

n is the unit normal; s denotes the tangential direction.

$$\frac{\partial \Pi_h}{\partial u_{s,i}} = -n_z \frac{\partial \Pi_h}{\partial u_{r,i}} + n_r \frac{\partial \Pi_h}{\partial u_{z,i}} = R_{s,i} = 0$$

3. Fields at a point in the element

$$\boldsymbol{u}(r,z) = \boldsymbol{N}\boldsymbol{u}_e, \qquad \boldsymbol{\varepsilon} = \boldsymbol{B}\boldsymbol{u}_e, \qquad \boldsymbol{\sigma} = \boldsymbol{D}\boldsymbol{\varepsilon}$$

After solving, reconstruct all displacement components before evaluating the fields.

Prescribed temperature and free thermal strain

1. Temperature relative to the stress-free reference state

$$T = T(r, z), \qquad \Delta T(r, z) = T(r, z) - T_0$$

T_0 is the stress-free reference temperature. Temperature differences in °C and K are numerically equal.
We do not solve the heat equation here. The temperature field is mapped onto the mechanical mesh.

2. Free expansion of an isotropic material

$$\ell = \ell_0(1 + \alpha_T \Delta T), \qquad \frac{\ell - \ell_0}{\ell_0} = \alpha_T \Delta T$$

ℓ_0 and ℓ are the lengths before and after heating. The constant coefficient of linear thermal expansion has units 1/K.
It is distinct from the heat transfer coefficient denoted by α in the thermal boundary-condition diagram.

3. Free thermal strain vector

$$\boldsymbol{\varepsilon}_{\text{th}} = \alpha_T \Delta T \begin{bmatrix} 1 & 1 & 1 & 0 \end{bmatrix}^{\mathsf{T}}$$

Component ordering: $(\varepsilon_{rr}, \varepsilon_{zz}, \varepsilon_{\varphi\varphi}, \gamma_{rz})$

The temperature field is known. We now account for its effect on the mechanical response.

Total, elastic and thermal strains

1. Kinematics gives total strain; Hooke's law relates stress to elastic strain

$$\boldsymbol{\varepsilon} = \boldsymbol{B}\boldsymbol{u}_e, \qquad \boldsymbol{\varepsilon}_{\text{el}} = \boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_{\text{th}}, \qquad \boldsymbol{\sigma} = \boldsymbol{D}\boldsymbol{\varepsilon}_{\text{el}}$$

The subscripts el and th denote the elastic and free thermal parts of the total strain ε.
The term 'thermal stress' identifies its cause; Hooke's law remains unchanged.

2. Free uniform heating

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_{\text{th}} \quad \Longrightarrow \quad \boldsymbol{\sigma} = 0$$

If the supports allow this expansion,
no stress develops.

A nonuniform temperature field may cause
incompatible free thermal expansion.

3. Axial expansion is prevented

$$\sigma_{zz} = E(\varepsilon_{zz} - \alpha_T \Delta T),$$
$$\varepsilon_{zz} = 0 \quad \Longrightarrow \quad \sigma_{zz} = -E\alpha_T \Delta T.$$

Uniaxial example: the lateral surfaces are
traction-free, so transverse stresses vanish.
This does not constrain all three normal strains.

Stress depends on elastic strain: the difference between the actual total strain and free thermal strain.

Thermal strain in the potential energy

1. Use elastic strain in the strain energy

$$\Pi_e(\boldsymbol{u}_e;T) = \frac{1}{2}\int_{\Omega_e} (\boldsymbol{B}\boldsymbol{u}_e - \boldsymbol{\varepsilon}_{\text{th}})^{\mathsf{T}} \boldsymbol{D}(\boldsymbol{B}\boldsymbol{u}_e - \boldsymbol{\varepsilon}_{\text{th}})\, 2\pi r\, dA - W_{b,e} - W_{t,e}$$

2. Expand the quadratic form for symmetric D

$$\Pi_e = \frac{1}{2}\boldsymbol{u}_e^{\mathsf{T}}\boldsymbol{K}_e\boldsymbol{u}_e - \int_{\Omega_e} (\boldsymbol{B}\boldsymbol{u}_e)^{\mathsf{T}}\boldsymbol{D}\boldsymbol{\varepsilon}_{\text{th}}\, 2\pi r\, dA - \boldsymbol{u}_e^{\mathsf{T}}(\boldsymbol{f}_e^b + \boldsymbol{f}_e^t) + C_e(T),$$

$$C_e(T) = \frac{1}{2}\int_{\Omega_e} \boldsymbol{\varepsilon}_{\text{th}}^{\mathsf{T}}\boldsymbol{D}\boldsymbol{\varepsilon}_{\text{th}}\, 2\pi r\, dA.$$

The quadratic mechanical term has already been derived. The cross term is linear in the nodal displacements. $C_e(T)$ depends on the prescribed temperature, but not on the unknown displacements.

Temperature adds a linear term and an energy term independent of the displacements.

The new linear term and thermal nodal loads

1. Multiply thermal strain by the constitutive matrix

$$\boldsymbol{D}\boldsymbol{\varepsilon}_{\text{th}} = \frac{E\alpha_T \Delta T}{1-2\nu}\begin{bmatrix}1 & 1 & 1 & 0\end{bmatrix}^{\mathsf{T}}$$

$$\Pi_{e,\text{mix}} = -\frac{2\pi E\alpha_T}{1-2\nu}\int_{\Omega_e} r\Delta T(\varepsilon_{rr}+\varepsilon_{zz}+\varepsilon_{\varphi\varphi})\,dA$$

2. Express strains in terms of nodal displacements

$$\Pi_{e,\text{mix}} = -\sum_{\ell\in\{i,j,m\}}\left(u_{r,\ell}f^{\text{th}}_{r,\ell}+u_{z,\ell}f^{\text{th}}_{z,\ell}\right) = -\boldsymbol{u}_e^{\mathsf{T}}\boldsymbol{f}^{\text{th}}_e$$

$$f^{\text{th}}_{r,\ell} = \frac{2\pi E\alpha_T}{1-2\nu}\left[\frac{b_\ell}{2A_e}\int_{\Omega_e} r\Delta T\,dA + \int_{\Omega_e} N_\ell \Delta T\,dA\right],$$

$$f^{\text{th}}_{z,\ell} = \frac{2\pi E\alpha_T}{1-2\nu}\frac{c_\ell}{2A_e}\int_{\Omega_e} r\Delta T\,dA.$$

ℓ = i, j, m. The coefficients $b\ell$, $c\ell$ and shape functions $N\ell$ are determined by the triangle geometry.

The thermal load vector arises from the energy cross term, not from an external pressure.

Exact integration of a linear temperature field

1. Temperature interpolation and geometric integrals

$$\Delta T = N_i \Delta T_i + N_j \Delta T_j + N_m \Delta T_m, \qquad r = N_i r_i + N_j r_j + N_m r_m$$

$$\int_{\Omega_e} N_\ell^2 \, dA = \frac{A_e}{6}, \qquad \int_{\Omega_e} N_\ell N_p \, dA = \frac{A_e}{12} \quad (\ell \neq p)$$

ℓ and p denote nodes i, j, m. ΔT_i, ΔT_j and ΔT_m are the prescribed nodal temperature changes.

2. Exact evaluation of the two integrals

$$\int_{\Omega_e} r \Delta T \, dA = \frac{A_e}{12} \left[(r_i + r_j + r_m)(\Delta T_i + \Delta T_j + \Delta T_m) + r_i \Delta T_i + r_j \Delta T_j + r_m \Delta T_m \right],$$

$$\int_{\Omega_e} N_\ell \Delta T \, dA = \frac{A_e}{12} (\Delta T_i + \Delta T_j + \Delta T_m + \Delta T_\ell).$$

Insert these integrals into the preceding slide's load expressions for constant elastic properties and expansion coefficient.
Exact thermal integration does not imply exact stiffness integration.

Both cross-term integrals are evaluated exactly for a temperature field linear over the triangle.

Constant temperature and the centroid approximation

1. ΔT is constant throughout the element

$$\int_{\Omega_e} r\Delta T\, dA = A_e r_c \Delta T, \qquad \int_{\Omega_e} N_\ell \Delta T\, dA = \frac{A_e \Delta T}{3}$$

$$f^{\text{th}}_{r,\ell} = \frac{2\pi E \alpha_T \Delta T A_e}{1-2\nu}\left(\frac{r_c b_\ell}{2A_e} + \frac{1}{3}\right), \qquad f^{\text{th}}_{z,\ell} = \frac{2\pi E \alpha_T \Delta T A_e}{1-2\nu}\frac{r_c c_\ell}{2A_e}$$

$r_c = (r_i + r_j + r_m)/3$: centroid radius, $\qquad \ell = i, j, m.$

2. One-point quadrature for a variable temperature field

$$\Delta T_c = \frac{\Delta T_i + \Delta T_j + \Delta T_m}{3},$$

$$\begin{cases} \displaystyle\int_{\Omega_e} r\Delta T\, dA \approx A_e r_c \Delta T_c, \\ \displaystyle\int_{\Omega_e} N_\ell \Delta T\, dA \approx A_e \Delta T_c/3. \end{cases}$$

Assess this approximation separately. For a linear field, use the exact integrals on Slide 27.

The constant-ΔT formulas are exact; replacing a variable field by its centroid value is an approximation.

Thermal-term derivatives and the assembled functional

1. Derivatives with respect to radial and axial nodal displacements

$$\frac{\partial \Pi_{e,\mathrm{mix}}}{\partial u_{r,\ell}} = -f^{\mathrm{th}}_{r,\ell}, \qquad \frac{\partial \Pi_{e,\mathrm{mix}}}{\partial u_{z,\ell}} = -f^{\mathrm{th}}_{z,\ell}, \qquad \frac{\partial C_e(T)}{\partial u_{r,\ell}} = \frac{\partial C_e(T)}{\partial u_{z,\ell}} = 0$$

$$\boldsymbol{f}^{\mathrm{th}}_e = \begin{bmatrix} f^{\mathrm{th}}_{r,i} & f^{\mathrm{th}}_{z,i} & f^{\mathrm{th}}_{r,j} & f^{\mathrm{th}}_{z,j} & f^{\mathrm{th}}_{r,m} & f^{\mathrm{th}}_{z,m} \end{bmatrix}^{\mathsf{T}}$$

2. Assemble thermal loads at shared nodes

$$F^{\mathrm{th}}_{I_a} \leftarrow F^{\mathrm{th}}_{I_a} + (f^{\mathrm{th}}_e)_a, \quad (I_1, \dots, I_6) = (2i-1, 2i, 2j-1, 2j, 2m-1, 2m)$$

Initialize the thermal load vector to zero. a = 1,…,6. Retain the mechanical load vector F.

3. Total global potential energy at a prescribed temperature

$$\Pi_h(\boldsymbol{u}; T) = \frac{1}{2}\boldsymbol{u}^{\mathsf{T}}\boldsymbol{K}\boldsymbol{u} - \boldsymbol{u}^{\mathsf{T}}(\boldsymbol{F} + \boldsymbol{F}^{\mathrm{th}}) + \sum_e C_e(T)$$

Minimize only this assembled functional over admissible independent displacements.

The new linear term adds a right-hand-side contribution; C(T) does not affect the displacements.

Thermoelastic system, constraints and solution

1. Stationarity of the assembled functional

$$\frac{\partial \Pi_h}{\partial u_{r,i}} = (\boldsymbol{K}\boldsymbol{u} - \boldsymbol{F} - \boldsymbol{F}^{\text{th}})_{2i-1} = 0, \qquad \frac{\partial \Pi_h}{\partial u_{z,i}} = (\boldsymbol{K}\boldsymbol{u} - \boldsymbol{F} - \boldsymbol{F}^{\text{th}})_{2i} = 0$$

These equations apply only to independent unknown components after imposing the constraints.

$$\boldsymbol{K} \begin{bmatrix} u_{r,1} \\ u_{z,1} \\ \vdots \\ u_{r,n} \\ u_{z,n} \end{bmatrix} = \begin{bmatrix} F_1 + F_1^{\text{th}} \\ F_2 + F_2^{\text{th}} \\ \vdots \\ F_{2n-1} + F_{2n-1}^{\text{th}} \\ F_{2n} + F_{2n}^{\text{th}} \end{bmatrix} + \boldsymbol{R}$$

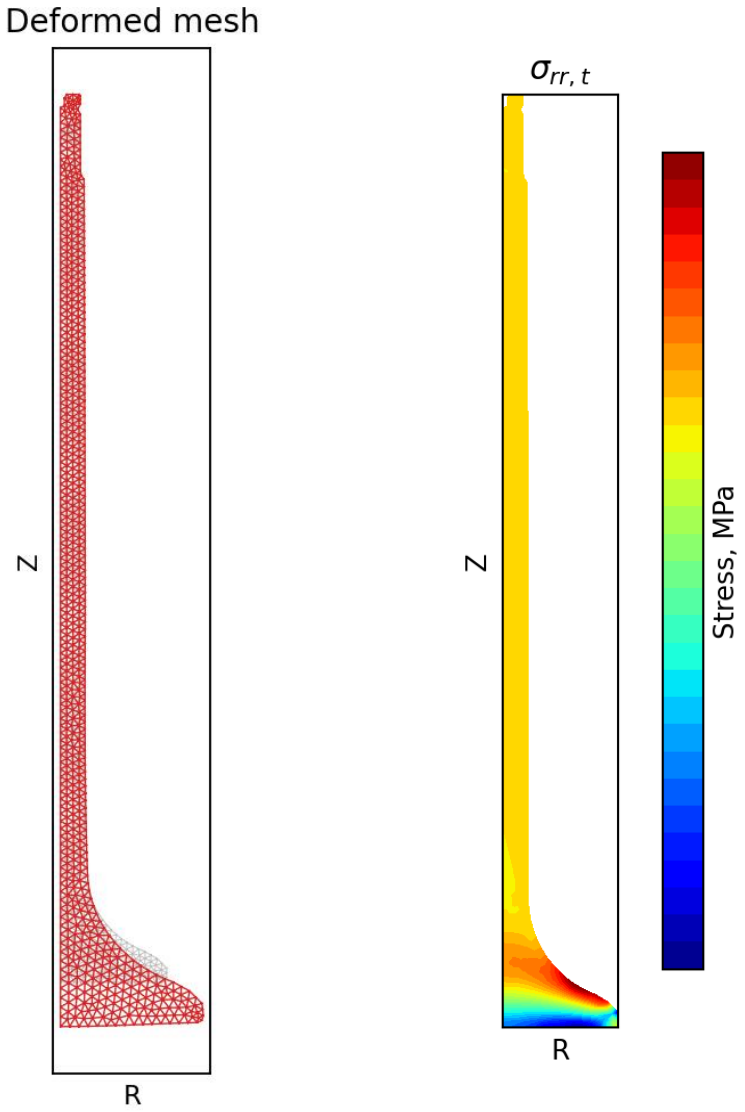
2. Impose prescribed displacements and solve

$$\sum_{b \in J} K_{ab} u_b = F_a + F_a^{\text{th}} - \sum_{c \in C} K_{ac} \bar{u}_c, \qquad a \in J$$

J and C index unknown and prescribed components. After solving, reconstruct the full vector u.
K is unchanged if E, ν, the geometry and the constraint model remain unchanged.

For an unchanged linear model, temperature modifies the right-hand side of the system.

Recovering thermoelastic fields and reactions



1. Strain and stress fields

$$\boldsymbol{\varepsilon}(r,z) = \boldsymbol{B}(r,z)\boldsymbol{u}_e,$$
$$\boldsymbol{\varepsilon}_{\mathrm{el}}(r,z) = \boldsymbol{\varepsilon}(r,z) - \boldsymbol{\varepsilon}_{\mathrm{th}}(r,z),$$
$$\boldsymbol{\sigma}(r,z) = \boldsymbol{D}\boldsymbol{\varepsilon}_{\mathrm{el}}(r,z).$$

Subtract thermal strain using the same T and T_0
as in the equivalent thermal nodal load vector.

2. Reactions from the original system

$$\boldsymbol{R} = \boldsymbol{K}\boldsymbol{u} - \boldsymbol{F} - \boldsymbol{F}^{\mathrm{th}}$$

3. Two necessary steps

The thermal load vector determines the equilibrium displacements. Subtracting thermal strain isolates the elastic part of the total strain.
This does not count temperature effects twice.

Total strain is determined by the displacements; stress is determined by its elastic part.



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Special Topics in Finite Element Modeling for IC engines

CONCLUSION